Linear Algebra

Midterm Examination
SSE, CUHK(SZ)

Nov 4, 2018

Name: _____ Student ID: _____

The exam contains 9 questions. Answer all the questions in the Answer Book.
This page has no questions, but a table of notations

Table of Notations

$\mathbb{R}$	the set of real numbers
	Without otherwise specified, all matrices have entries from $\mathbb{R}$.
$\text{Null}(A)$	the null space of a matrix A
$\dim(V)$	the dimension of a vector space V
A^T	the transpose of matrix A
$\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$	the linear span of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_r$

This study source was downloaded by 100000886444315 from CourseHero.com on 06-20-2024 11:40:47 GMT -05:00

https://www.coursehero.com/file/222652422/MAT2040-Midterm-2018v2pdf/

I Midterm Questions

1. (12 points) Consider the following system of linear equations with x, y, z as the variables.

$$\begin{cases} x + 4y - 2z = 1 \\ x + 7y - 5z = 7 \\ 3y + qz = t \end{cases}$$

Find the values of q and t such that the system has:

(a) (4 points) Exactly one solution.

(b) (4 points) Infinitely many solutions.

(c) (4 points) Give the solution set in parametric vector form when the system has infinitely many solutions.

2. (13 points) Let $A = \begin{bmatrix} 5 & -3 & 4 & 1 \\ -15 & 12 & -13 & -6 \\ -5 & 9 & -5 & -9 \\ -10 & 9 & -7 & -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ -9 \\ -14 \\ 0 \end{bmatrix}$.

(a) (8 points) Find the LU decomposition of A.

(b) (5 points) Solve $A\mathbf{x} = \mathbf{b}$ by using $L\mathbf{y} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$.

3. (8 points) Let A be a 5×4 matrix, and let $\mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ -5 \\ 1 \\ 10 \end{bmatrix}$. Suppose the solution set for $A\mathbf{x} = \mathbf{b}$ is

$$\{(s_1, s_2, s_3, s_4) : (s_1, s_2, s_3, s_4) = (4 + 2t, -2 + t, 1 - t, 10) \text{ for } t \in \mathbb{R}\}.$$

Give the solution set for $A\mathbf{x} = \mathbf{0}$ in parametric vector form.

4. (8 points) Suppose B is a 4×3 matrix with null space $\text{Span}\left\{\begin{bmatrix} 1 \\ -4 \\ 4 \end{bmatrix}\right\}$, and let

$$C = \begin{bmatrix} \mathbf{b}_1, & \mathbf{b}_2, & \mathbf{b}_3, & 2\mathbf{b}_1 + 5\mathbf{b}_3 \end{bmatrix},$$

where $\mathbf{b}_i$ is the i-th column of B for $i = 1, 2, 3$.

Give the solution set for $C\mathbf{x} = \mathbf{0}$ (in any form). Clearly show your reasoning.

This study source was downloaded by 100000886444315 from CourseHero.com on 06-20-2024 11:40:47 GMT -05:00

https://www.coursehero.com/file/222652422/MAT2040-Midterm-2018v2pdf/

5. (10 points) Consider the following vectors

$$\mathbf{x}_1 = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \qquad \mathbf{x}_2 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}, \qquad \mathbf{x}_3 = \begin{bmatrix} 1 \\ -12 \\ -6 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} -9 \\ -2 \\ 5 \end{bmatrix}.$$

   (a) (4 points) Is $\mathbf{b} \in \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$?

   (b) (6 points) Does $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ form a basis for $\mathbb{R}^3$?

6. (18 points) Judge each of the following statements is true or false in general for a vector space V: If false, give a counter example, if true write a proof.

   (a) (4 points) If vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ in V are linearly dependent, then $\mathbf{v}_1$ can be written as a linear combination of $\mathbf{v}_2, \ldots, \mathbf{v}_r$.

   (b) (4 points) If vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in V are pairwise linearly independent, i.e., any two of them are linearly independent, then $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent.

   (c) (4 points) If there exist scalars $\alpha_1, \alpha_2, \cdots, \alpha_m$, not all zero, such that,

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_m\mathbf{v}_m + \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_m\mathbf{u}_m = \mathbf{0},$$

   then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ in V are linearly dependent, and $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ in V are also linearly dependent.

   (d) (6 points) If W, U are subspace of V, then $W \cap U$ is a subspace of V.

7. (6 points) Let A be a square $n \times n$ matrix over real numbers. Find all the statements that are equivalent to "A is nonsingular (invertible)".

   A. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.

   B. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$.

   C. A is row equivalent to I.

   D. A is a product of a sequence of finite elementary matrices.

   E. The rows of A are linearly independent.

   F. The columns of A span $\mathbb{R}^n$.

   G. $\dim(\text{Null}(A)) = 0$.

8. (10 points) Find a basis for each of the following vector spaces.

   (a) (5 points) The vector space of 3×3 symmetric matrices (i.e., those matrices satisfying $A = A^T$).

   (b) (5 points) The space of all polynomials of the form $ax^2 + bx + 2a$, where $a, b \in \mathbb{R}$ and x is the variable.

This study source was downloaded by 100000886444315 from CourseHero.com on 06-20-2024 11:40:47 GMT -05:00

https://www.coursehero.com/file/222652422/MAT2040-Midterm-2018v2pdf/

9. (15 points) Let A be an $m \times n$ matrix and let B be an $m \times m$ invertible matrix.
   (a) (7 points) Show that $\text{Null}(BA) = \text{Null}(A)$.
   (b) (8 points) Show that $(BA)^T$ and A^T have the same column space.

This study source was downloaded by 100000886444315 from CourseHero.com on 06-20-2024 11:40:47 GMT -05:00

https://www.coursehero.com/file/222652422/MAT2040-Midterm-2018v2pdf/